

Thermodynamics

Introduction to thermodynamics - Energy transfer heat, work and mass.

Thermodynamics:-

Deals with the transfer of energy from one form to another or from one place to another.

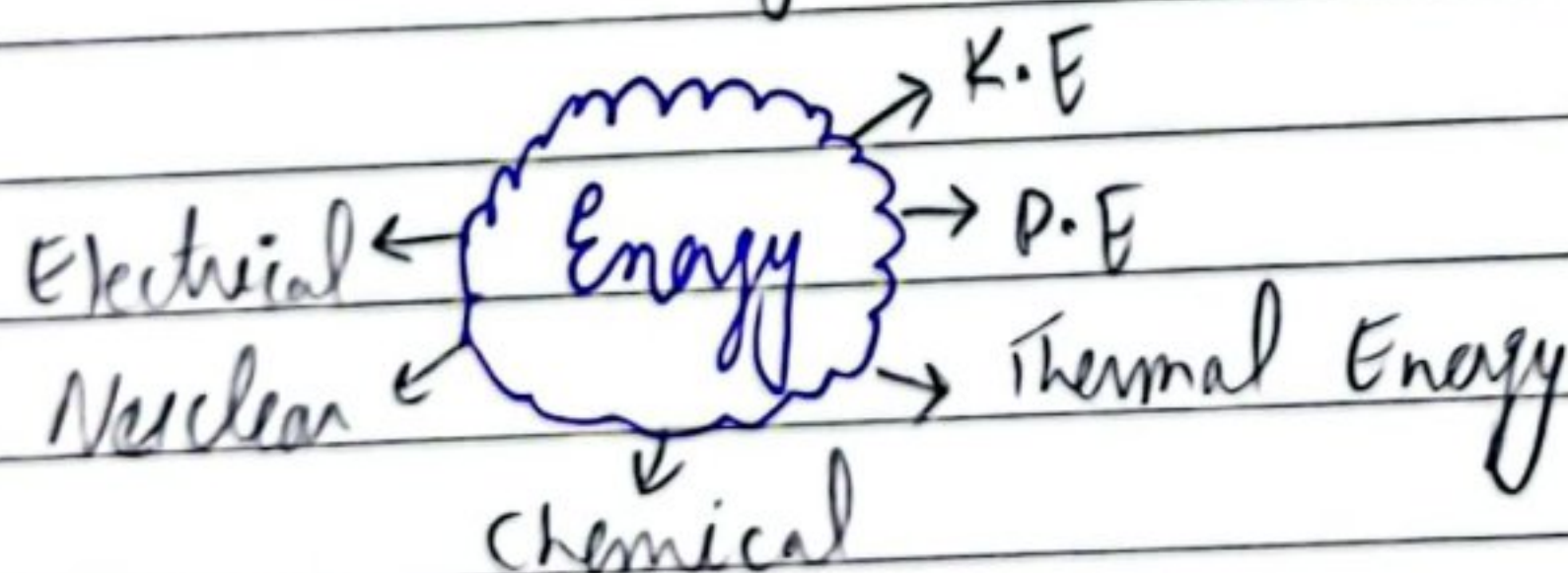
Energy:-

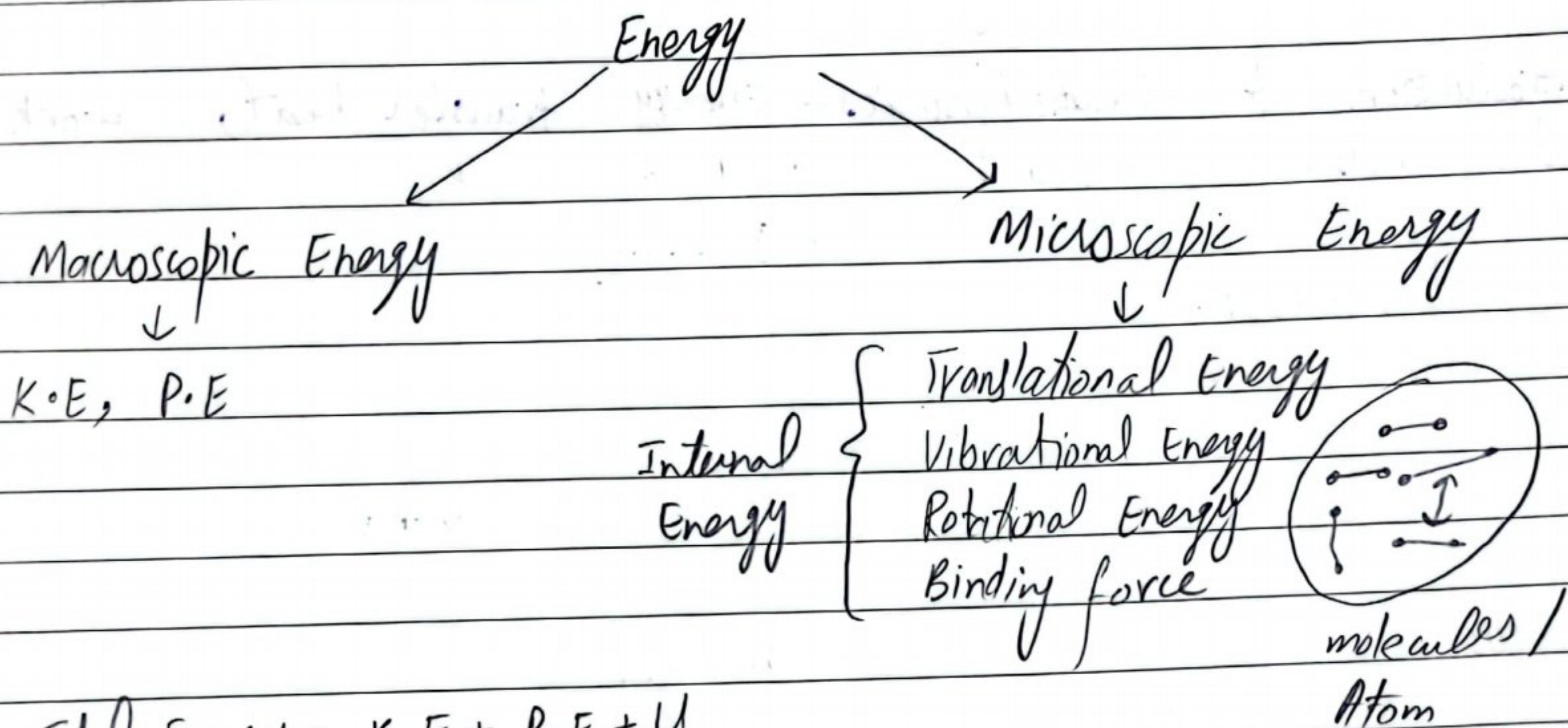
A quantitative property that must be transferred from a body or physical system to perform work or transfer heat.

Heat:

Form of energy that is transferred between two bodies due to difference in the temperature.

↳ Rate of movement of molecules



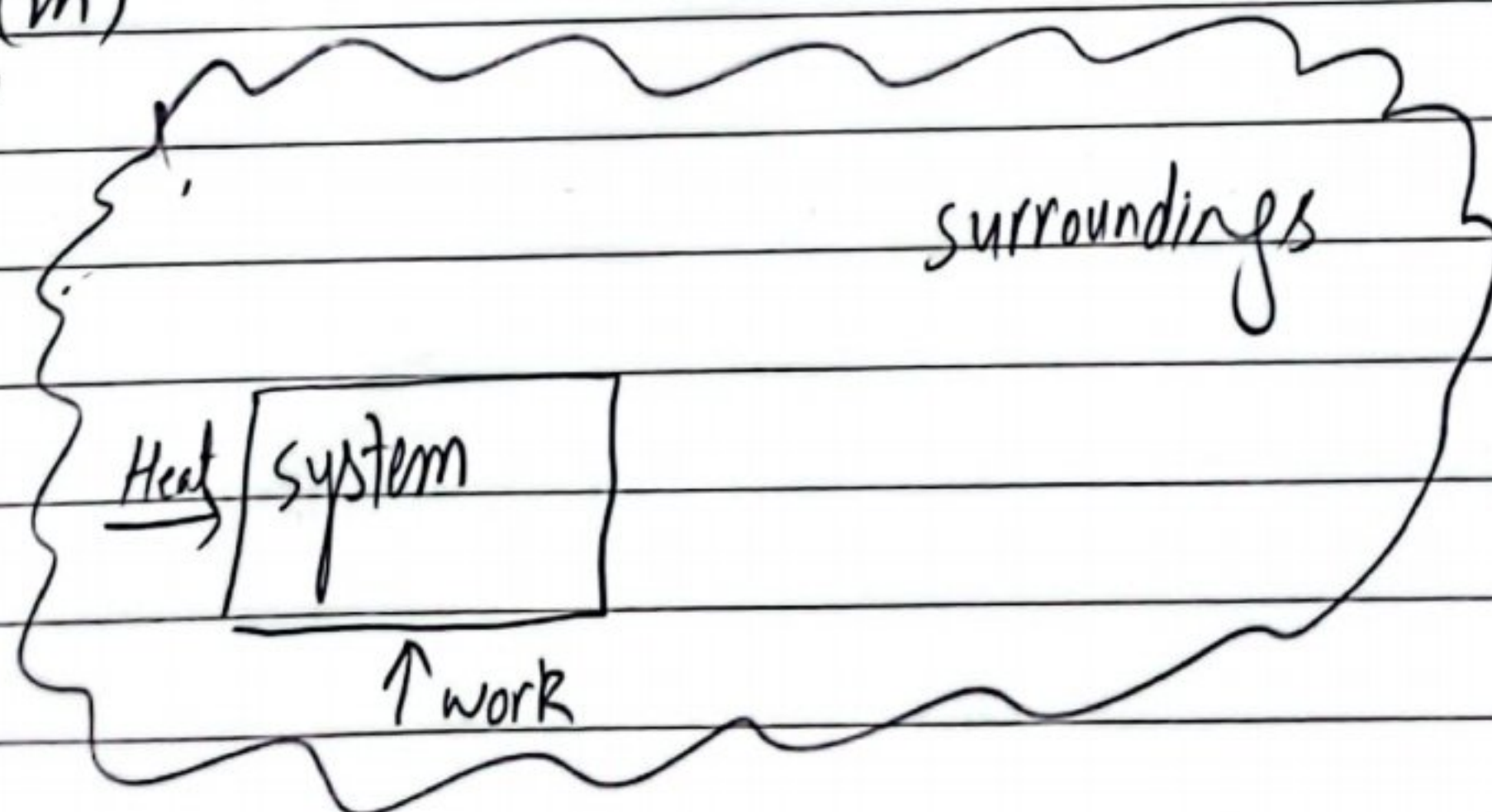


$$\text{Total Energy} = K.E + P.E + U$$

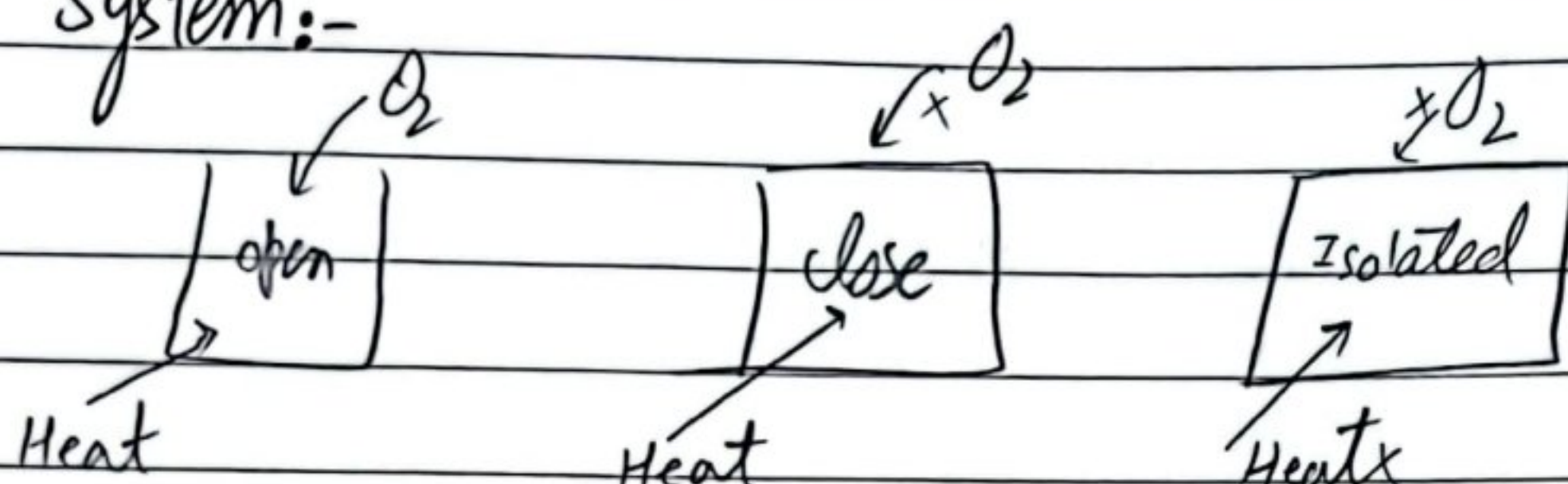
$$= \frac{1}{2}mv^2 + mgh + U$$

Mechanism of Energy transfer

- 1- Work Done (W) → Joule → Mechanical energy
- 2- Heat (q) → Joule → Thermal Energy
- 3- Mass (m) flow



System:-



~~$\Delta U = q + w$ (Chemistry)~~

$\Delta U = q - w$ (Physics) → use this in our course

Chemistry:-

~~$W = -ve$ → Work done by the system~~

~~$W = +ve$ → Work done on the system~~

Physics:-

$W \rightarrow -ve$ → Work done ON the system

$W \rightarrow +ve$ → Work done BY the system

Heat:-

$q \rightarrow +ve$ (absorb) → endothermic
 $q \rightarrow -ve$ (release) → exothermic

Example:-

① 100 J of work is done on the system
 U increases by 74 J. How much energy is transferred as heat?

$\Delta U = q - w$

$-74 = q - 100$

$q = -26 \text{ J}$

② A sample of gas does 150 J of work against its surrounding and loses 90 J of internal energy in the process. Does the gas lose or gain heat?

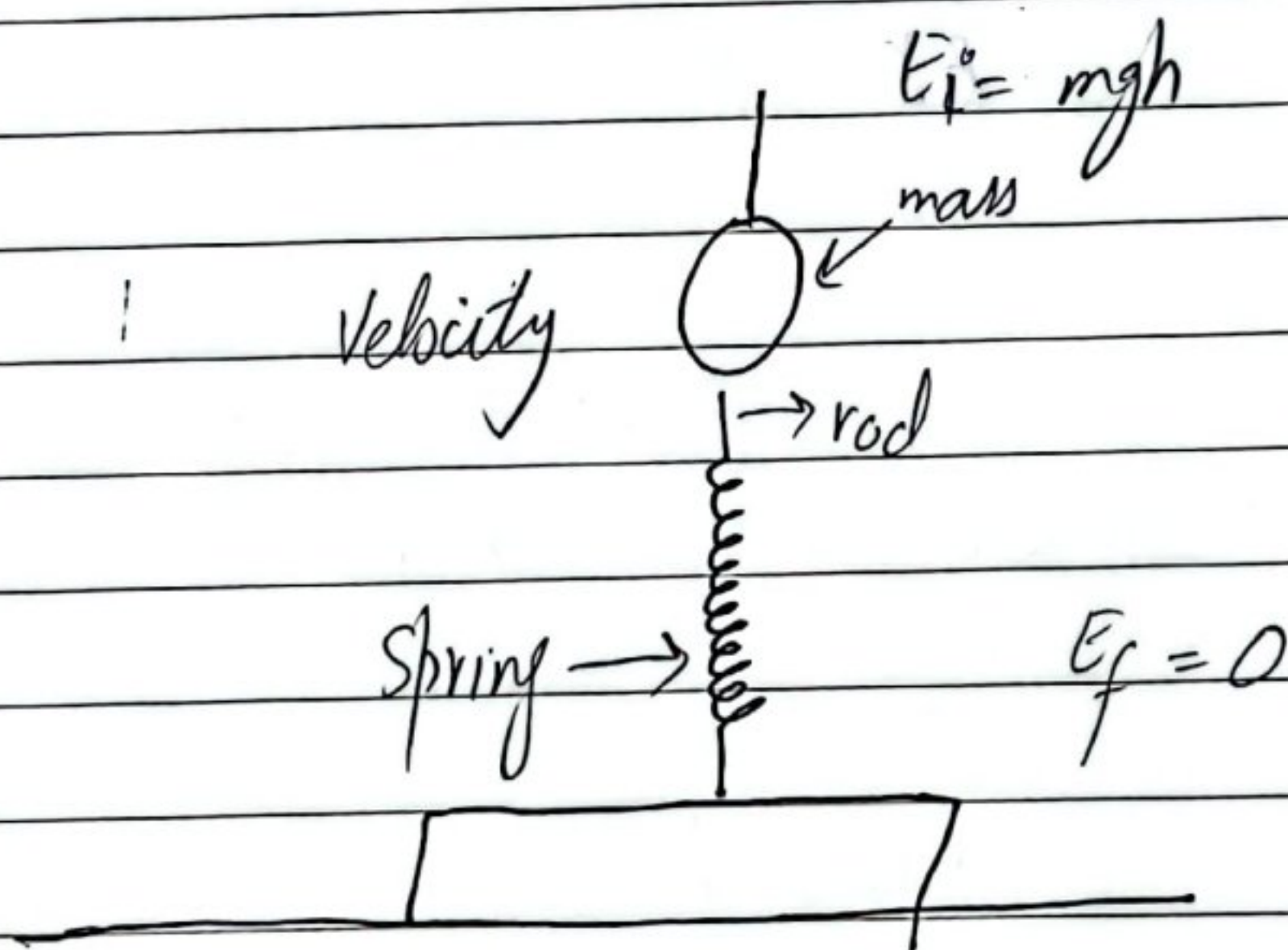
$$\Delta U = q - w$$

$$q = \Delta U + w$$

$$q = -90 + 150$$

$$q = +60 \text{ J}$$

③



Properties of Pure substance

Pure substance :-

A pure substance is defined as a material that has a homogeneous and invariable chemical composition.

Homogeneity

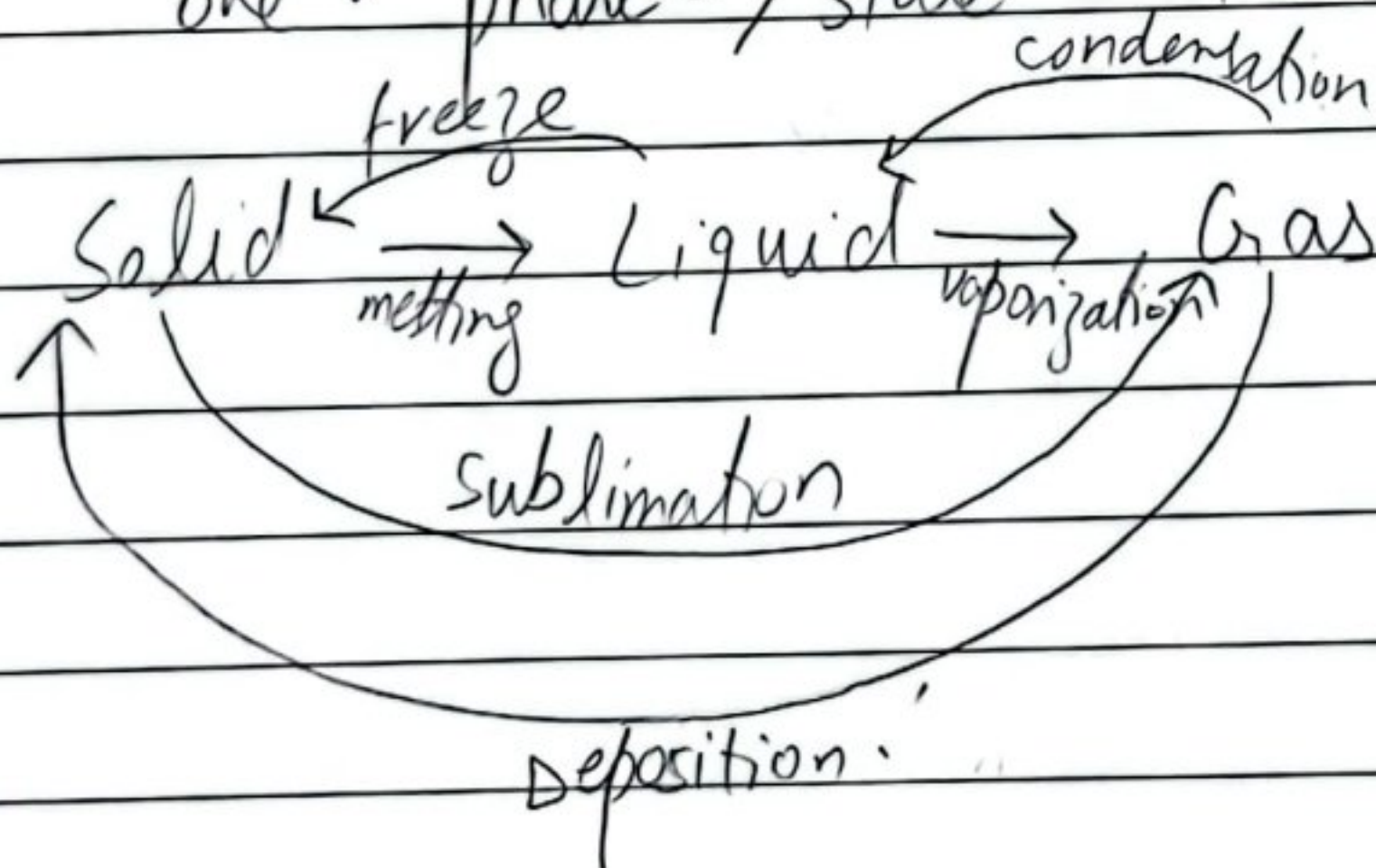
The substance is the same throughout, with no regions having a different composition.

Invariable composition.

The chemical composition doesn't change even when the substance transitions between different phases.

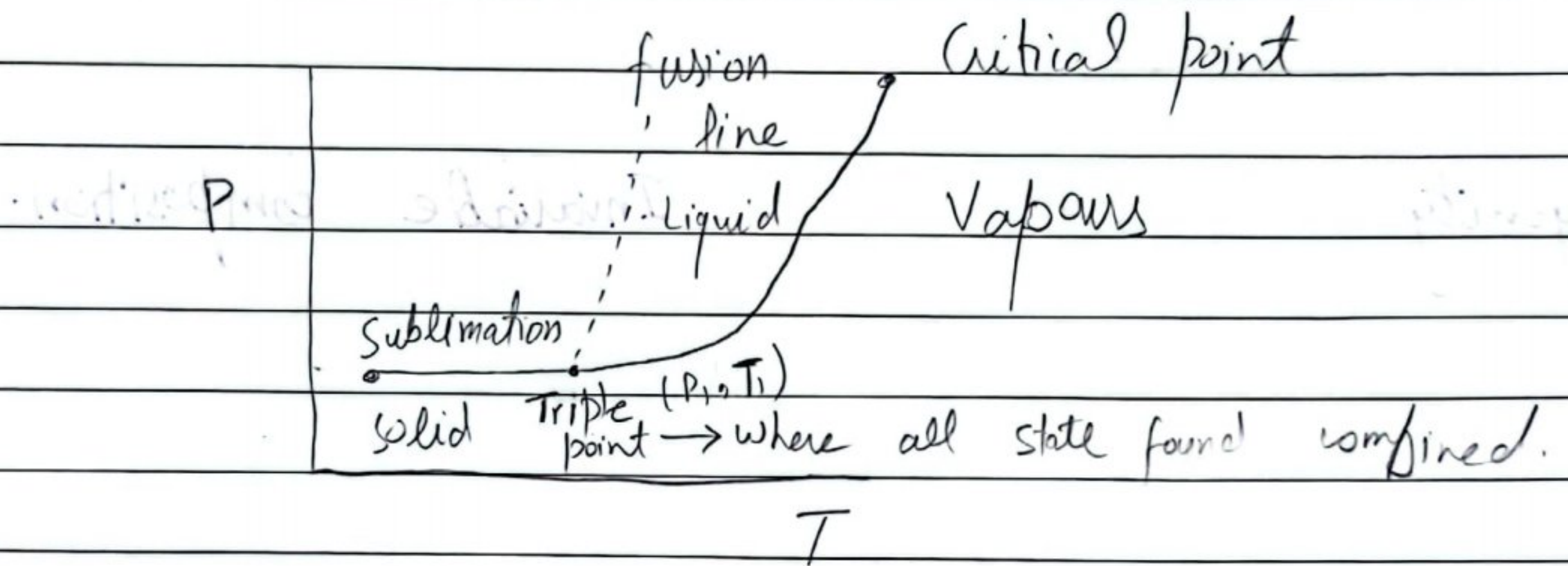
Phase Change and Diagram

Phase change of material occurs when it transitions from one phase/state to another.



Saturation Temperature :-

The specific temperature at which a substance undergoes a phase change under a given pressure. (vaporization, condensation)

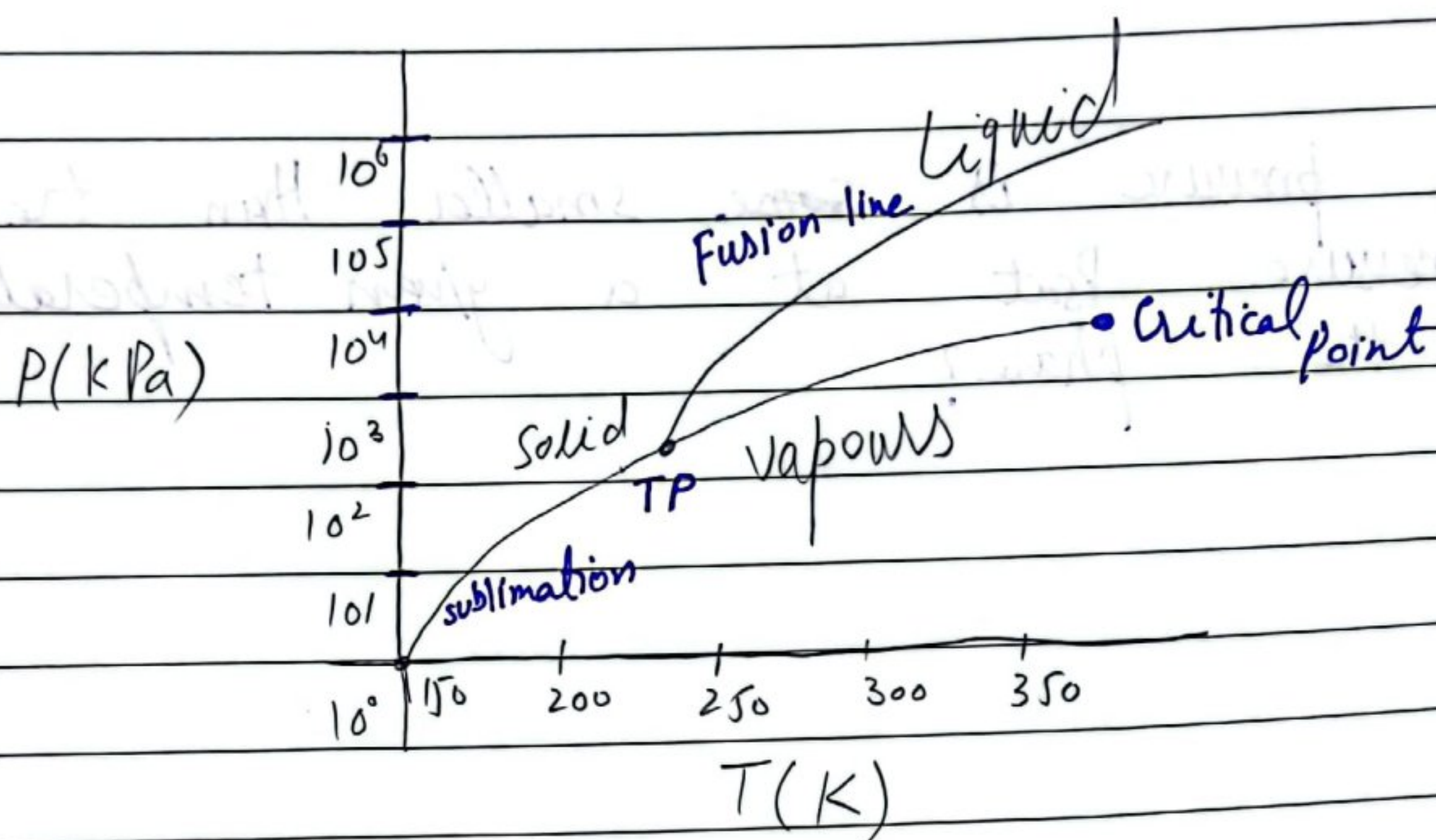


Triple point of water occurs at 0.01°C , 0.6117 kPa

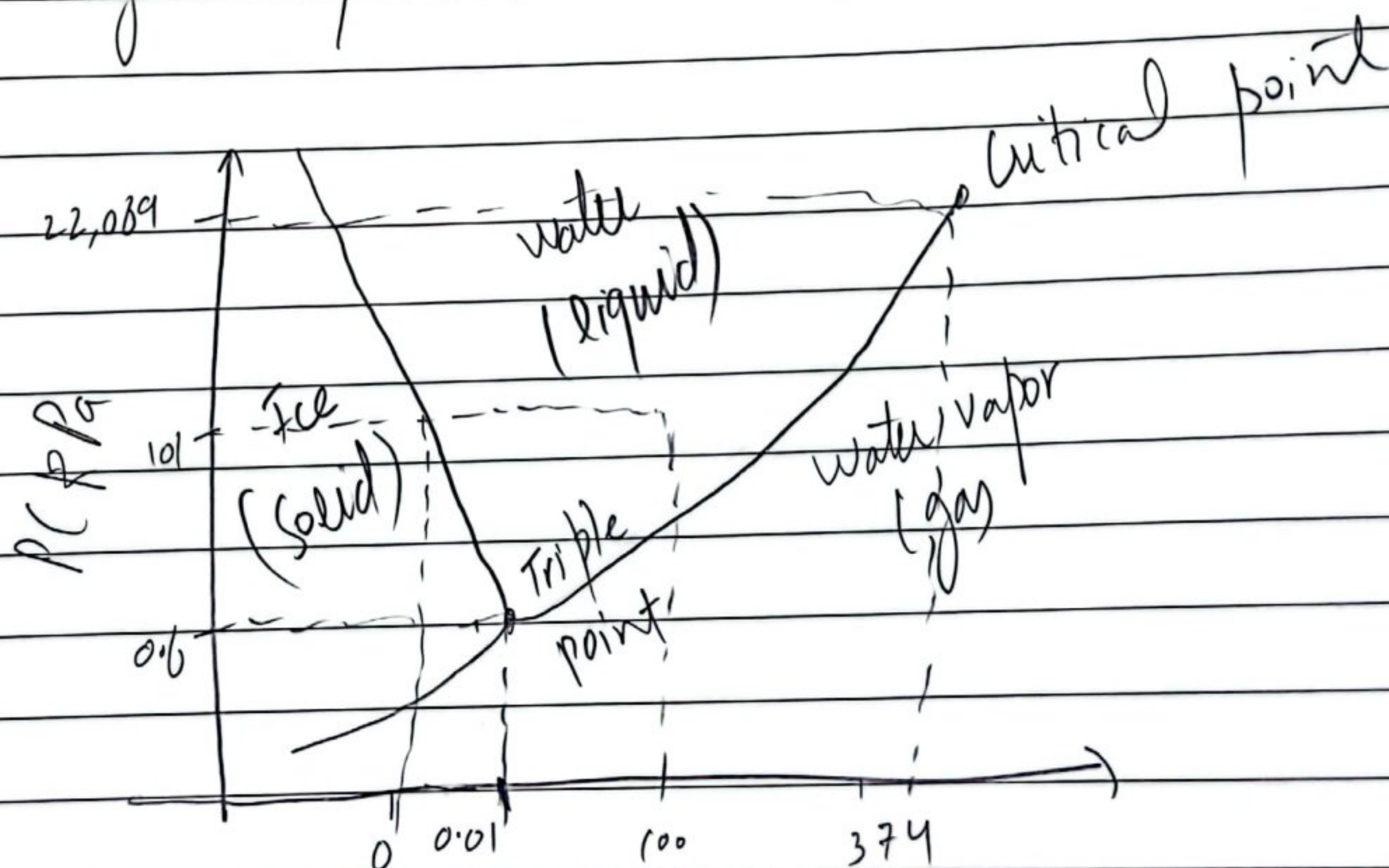
Critical Point marks the end of vaporization curve on a phase diagram. Beyond this point liquid & gaseous phase becomes indistinguishable.

Critical Point of water 374.14°C and 22.09 MPa

Phase diagram of CO_2



Phase diagram of H_2O



Question: what is the lowest temperature at which water can remain in the liquid form?

0.01°C

Question: If the pressure is ~~same~~ smaller than the smallest pressure P_{sat} at a given temperature T , what is the phase?

Question: what is the lowest temperature at which water can remain in the liquid form?

0.01°C

Question: If the pressure is ~~some~~ smaller than the smallest pressure P_{sat} at a given temperature T , what is the phase?

Day 4

The P-v-T surface:-

- critical for understanding how substances behave under different conditions. (P, v, T)
- Applications → Refrigeration, power generation, etc. compressor etc.

Refer to figure 2.6 from book

P-v-T surface:-

- Captures the relationship b/w the pressure, volume and temperature

P-v-T surface of water:-

- Ice → has larger volume than liquid state.
- liquid water
(water has abnormal behaviour)

The fusion line (Solid-liquid) line on pressure-Temp diagram has a negative slope indicating the increasing of the pressure lowers the freezing point.

P-v-T surface of a typical substance:-

- shrinks on freezing
- Solid phase have lower volume than liquid phase.

The fusion line (Solid-liquid) line on pressure-Temp diagram has a positive slope indicating the increasing the pressure the freezing point.

Phase Diagram:-

- P-T diagram
- Fusion line
- Sublimation line
- Triple Point
- Critical Point

T-V diagram:-

- Show how volume changes with temperature at constant pressure illustrating phase change from liquid to vapour.
- Depicts the boiling phase.

P-V diagram:-

- Shows how pressure changes with specific volume at constant temperature.

Independent properties:-

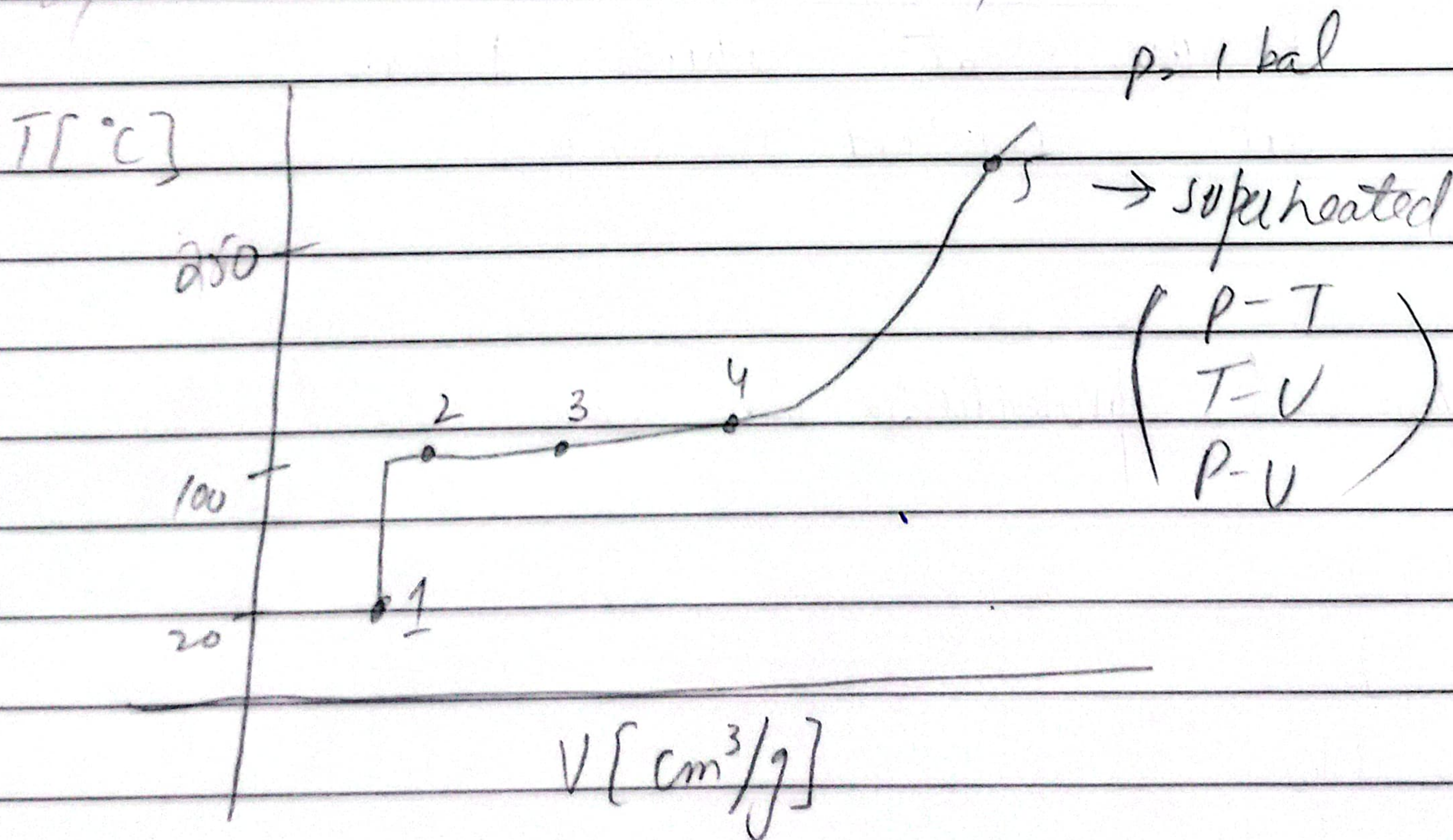
The states of substance can be defined by two independent properties. However, in the two phase region (during boiling or condensation), pressure and temperature are independent because they are fixed at a saturated point for a given pressure-temperature pair.

Tables of Thermodynamics properties

State 1 (liquid)	State 2 (liquid)	State 3 (liquid/vapour)	$P = 1 \text{ bar}$ $T = 100^\circ\text{C}$
$P = 1 \text{ bar}$	$P = 1 \text{ bar}$	$\rightarrow$ vapours	
$T = 20^\circ\text{C}$	$T = 100^\circ\text{C}$	$\rightarrow$ liquid	

Heat

State 4 (vapours)	State 5 (vapours)
$P = 1 \text{ bar}$	$P = 1 \text{ bar}$
$T = 100^\circ\text{C}$	$T = 250^\circ\text{C}$



Purpose:-

Thermodynamics tables provide essential data for various substance that describe their thermodynamic properties (like temperature, pressure, volume, specific volume, internal energy like enthalpy and entropy)

Saturation temperature:-

Temperature at which phase change occurs at constant Pressure

Saturation Pressure:-

Pressure at which phase change occurs at constant Temperature.

⇒ with 2 independent properties, we can find the rest

- Temperature (T)
- Pressure (P)
- Specific volume (v)
- Specific Internal energy (u)
- Specific Enthalpy (h)
- Specific entropy (s)

- Linear Interpolation
- Polynomial Regression
- Moving Average
- Lagrange Interpolation.

Saturated liquid and Vapour tables

Table B1.1:-

V_f = Specific volume of the saturation liquid
 V_g = Specific volume of the saturation vapour
 V_{fg} :- Difference between V_g & V_f

$$V_{fg} = V_g - V_f$$

First Column:- Temperature

Second Column:- Corresponding Saturation Pressure (P_{sat})

Table B 1.2:-

- contains the same information as B 1.1 but organized pressure
- Useful